

AI Detector report by

Model v7.1.0 

4,100 words

9/14/2026

12%

of text is likely AI

AI-generated  12%Human-written & AI-refined  0%Human-written  88%

Caution: Our AI Detector is advanced, but no detectors are 100% reliable, no matter what their accuracy scores claim. Never use AI detection alone to make decisions that could impact a person's career or academic standing.

Introduction

Designs for Equinix Consulting Group was selected as the lead data center architecture consultant to design and engineer a flexible, highly reliable, next generation data center infrastructure for eSewa Ltd., the oldest and first individual Nepalese licensed (by Nepal Rastra Bank (NRB)) Payment Service Provider (PSP) and ISO 27001:2022 certified digital financial network. eSewa provides wallet-to-wallet transfers, merchant QR code transactions, bill payments, inter-bank clearing for all 77 districts of Nepal for over 7 million users and 500,000+ business outlets.

With digital financial inclusion extending as defined by "Digital Nepal Framework," the present infrastructure of eSewa is expected to see a exponential growth of transaction numbers, upcoming seasonal nigh traffic in the runup to major national festivals and compliance requirements of existing regulatory uptime. Equinix has been assigned to design an open profile enterprise grade data center at Kathmandu Valley Tech Corridor, that provides 100% financial clearing capability, zero data loss and low-latency payment processing.

EQUINIX CONSULTING GROUP

Data Center Infrastructure Design Project

Client

eSewa Ltd. (F1Soft Group)

Industry

Fintech / Digital Payment Gateway & Ecosystem

Location

Kathmandu Valley Tech Corridor, Nepal

Targeting Standard

Uptime Institute Tier III Concurrent Maintainability

Physical Layout

TIA-942 Compliant Telecommunications Spatial Zoning

Business Need

There is effectively no room for un-coordinated time delays or latency bottlenecks in financial operations. A short time outage at a grocery store counter, hospital checkout counter or on a payment gateway at a bank results in billions of failed financial settlements, customer rage and regulatory penalties imposed by Nepal Rastra Bank.

Our design adapts to the weaknesses of conventional perimeter-only server rooms by providing a contemporary Tier III Concurrently Maintainable data center architecture, able to perform maintenance, update to firmware or perform emergency repairs to any power line, cooling unit or network switch without shutting down the live payment databases or API gateways.

Core Engineering Objectives

The general aims for the eSewa Data Center Project can be defined under five operational pillars:

CORE PROJECT OBJECTIVE

1

High Availability

99.982% Site Uptime (Concurrent Maintainability)

2

Transaction Speed

Internal Latency < 50ms & API Processing Speed

3

Regulatory Security

PCI-DSS v4.0 Compliance & Zero-Trust Segmentation

4

Elastic Scalability

10x On-Demand Compute Bursting & Modular Pod Layout

5

Operational Efficiency

Target PUE <= 1.35 with Integrated DCIM (BMS/EMS)

1. High Availability & Operational Resilience

- Achieve **Uptime Institute Tier III Certification**, guaranteeing **99.982% annual operational uptime** (maximum allowable unplanned downtime of hours per year).

- Eliminate single points of failure (SPOF) across power distribution paths, HVAC cooling loops, and high-speed network topologies by implementing capacity redundancy and dual-path distribution (power feeds).

2. Low Latency & High-Throughput Transaction Processing

- Maintain an internal network latency of for inter-service microservice calls between API gateways, fraud-check engines, and core ledger databases.
- Implement a 2-Tier Spine-and-Leaf network fabric operating on high-speed backbones to handle extreme transactional concurrency during peak festive shopping periods (*Dashain* and *Tihar*).

3. Strict Compliance & Zero-Trust Security

- Ensure full physical and network compliance with **PCI-DSS v4.0** (Payment Card Industry Data Security Standard) and **ISO/IEC 27001** information security standards.
- Establish software-defined network micro-segmentation using **VXLAN-EVPN** technology to isolate payment card databases, public API gateways, and enterprise administrative traffic into separate, secure network zones.

4. Scalable Space & IT Hardware Infrastructure

- Deploy a modular floor layout compliant with **TIA-942 spatial standards**, designating clear zones for the Main Distribution Area (MDA), Horizontal Distribution Area (HDA), and Equipment Distribution Area (EDA).
- Provide physical and logical expansion capacity allowing the server rack footprint to expand from an initial baseline of 50 racks to 200 racks without requiring structural modifications or service interruptions.

5. Energy Efficiency & Intelligent DCIM Monitoring

- Maintain a target Power Usage Effectiveness (PUE) metric of , calculated as:
- Deploy a unified Data Center Infrastructure Management (DCIM) platform integrating a **Building Management System (BMS)** for environmental and security oversight with an **Energy Management System (EMS)** for real-time power analytics and PUE optimization.

2. Data Center Scenario and Business Concept

2.1 Target Organization Profile and Global Location Selection

- **Enterprise Profile:** The organization under study was eSewa Ltd which is a subsidiary of F1Soft Group and was the first licensed PSP (Payment Service Provider) in Nepal. eSewa would provide the infrastructure to the payment ecosystem of Nepal with thousands of consumers and traders using their service eSewa. It has a foothold of over 750000 active consumers, 50 plus banking and financial institutions and 500000 plus merchant outlets.
- **Global/Regional Location: Kathmandu Valley Tech Corridor, Nepal** (Primary Site), with active-passive disaster recovery mapping to Pokhara.
- **Site Justification:**
 - **Proximity to Financial Hubs:** The Nation's Bank, the Nepal Rastra Bank (NRB) and the commercially banks are mainly located in Kathmandu. This helps to reduce the minimum network-hop delay when interbank fund transfer is made through computerized funds transfer system (CFTS), as well.
 - **Fiber Backbone Access:** Direct integration with the Nepal Internet Exchange (NPIX) and cross-border redundant fiber-optic routes connecting through India and China.

2.2 Core Business Model Analysis

2.2.1 Internal Enterprise Service Infrastructure

The open payment clearing system of eSewa conducts real-time micro-payments, P2P transfers, utility payments and government revenue collection. The back-end technology architecture of eSewa provides digital ledgers, ACID-compliance databases, real-time fraud detection engines as well as customers' KYC processing services.

2.2.2 Public/Private Cloud Service Architecture

To comply with tough monitoring and reporting regulations and provide elastic scalability, eSewa implements the Hybrid Private Cloud Architecture on:

2.3 Long-Term Business Strategy and Market Value Proposition

- **Market Context:** Digitalization of payment in Nepal is in fast pace as a result of central bank cashless initiatives and increasing mobile penetration.
- **Peak Load Surge Dynamics:** Major festivals such as Dashain and Tihar have been shown to lead to 5x–10x the normal daily Transaction levels. The infrastructure must be able to elastically scale compute capacity without downtime to payment services.

- **Value Proposition:** Ensures the sub-50ms APIs response time, no transactions are lost, and the financial clearing continues 24/7 confidence was earned by all customers and the flagship position of eSewa is secured.

2.4 Industry Case Studies and Benchmarks

This design adopts architectural strategies proven by major global financial technology leaders:

Industry Benchmark

Architectural Strategy Applied

Direct Benefit to eSewa

Paytm (India)

Shifted from legacy 3-tier networks to flat **Spine-and-Leaf Clos topologies**.

Eliminates bandwidth bottlenecks and provides sub-second processing for millions of concurrent QR code payment requests.

GrabPay (Southeast Asia)

Implemented **Software-Defined Micro-segmentation** (VXLAN-EVPN).

Strictly isolates core ledger databases from public web gateways, ensuring PCI-DSS v4.0 compliance and preventing lateral malware movement.

3. Technical Goals and Operational Strategy Guidelines

3.1 Overview and Strategic Alignment

The data center should be designed to meet a specific set of technical objectives and operational approaches to help eSewa Ltd. to sustain its leading position in Nepal as an finest fintech and digital payment infrastructure provider. It is from the business requirements that define the technical approach to a set of engineering requirements that represent some concrete level of specifications at the technical level.

TECHNICAL GOALS & BUSINESS ALIGNMENT

High Availability (99.982%)

Prevents transaction loss during payment

Sub-50ms Latency

Guarantees instant QR/API processing

Elastic Scalability (10x)

Handles peak festive traffic (Dashain/Tihar)

Zero-Trust Security

Protects financial data & PCI-DSS v4.0

3.2 Deep Dive into Core Technical Goals

Goal 1: High Availability and Concurrent Maintainability (99.982% Uptime)

- **Engineering Requirement:** The data center infrastructure must provide 99.982% operational availability per annum, in full compliance with the standards defined by the Uptime Institute Tier III classification.
- **Mathematical Justification:**

System availability is defined by Mean Time Between Failures () and Mean Time To Repair ():

At **99.982% uptime**, the maximum allowable unplanned operational downtime is calculated as:

- **Operational Relevance to eSewa:** Processing of payments occurs 24 hours a day. Premium banking interfaces or a simple offline upgrade of a banking or wallet interface for a routine firmware upgrade or UPS battery replacement causes point-of-sale (POS) transaction failures on merchant networks. It is essential to obtain Tier III Concurrent Maintainability where any one power feed, cooling unit or switch can be maintained without affecting operational and live financial ledger processes.

Goal 2: Ultra-Low Transaction Latency and Sub-50ms API Processing

- **Engineering Requirement:** End-to-end network latency inside the data center fabric must be kept under **2ms**, while maintaining an overall transaction processing API round-trip time of .
- **Operational Relevance to eSewa:** In a single checkout, the Digital Payment Gateway interacts with several external service providers simultaneously: user authentication, risk/fraud screening, wallet debiting, core ledger logging, and clearing via bank gateway. The accumulated latency across these micro services are leads to timeout of the payment system. Using non blocking 100“GbE”/400“ GbE” Spine-and-Leaf switching networks and the NVMe storage cluster, the data transmission delay is minimized.

Goal 3: Elastic Scalability and Modular Capacity Bursting

- **Engineering Requirement:** The infrastructure needs to be designed to support ready horizontal scaling. It should be capable of absorbing 10x base transaction loads in the time span of several minutes, on demand, with possible re-structure of existing n-tier service delivery or service degradation.
- **Operational Relevance to eSewa:** Transaction volume fluctuates wildly in Nepal. During the two Arhaj festivals (Dashain and Tihar), when the utility bills are due towards the end of the month, the volume of transaction requests increases by up to 5. During these times, a modular rack architecture enables eSewa to run idle virtualized compute nodes.

Goal 4: Zero-Trust Security Architecture and Multi-Tier Compliance

- **Engineering Requirement:** Implement raw, software-defined micro-segmentation, hardware-led encryption, and hardware security module (HSM) isolation that fully meets the requirements of PCI-DSS v4.0, ISO 27001 and NRB IT Guidelines.
- **Operational Relevance to eSewa:** For a financial institution handling daily millions of private wallet PINs, OTP tokens, bank account numbers and such, perimeter security is not enough. Zero-trust means that no traffic within the internal network, all internal cross-zone API requests between the public gateway and the financial ledgers are seen as “untrusted”.

3.3 Operational Strategy Guidelines

The following is the set of operational strategy guidelines used to guide the data center life cycle in translating the four core technical goals into design and implementation:

Strategy A: Modular Physical & Logical Pod Design

Strategy B: Infrastructure as Code (IaC) & Continuous Automation

Strategy C: Energy Efficiency & Power Usage Effectiveness (PUE) Optimization

Strategy D: Concurrent Maintainability & Predictive Maintenance

Strategy A: Modular Physical & Logical Pod Deployment

- **Standardized Pod Framework:** Instead of deploying servers in an ad-hoc manner, compute, network, and storage hardware are deployed in standardized "Pods" (e.g., 10-rack standardized blocks).
- **Benefit:** Facilitates for useable power flow, localized cooling and rapid deploying. When eSewa needs more capacity a new pre-tested Pod is brought online, that does not affect current operation domains.

Strategy B: Infrastructure as Code (IaC) and Automation

- **Automated Configuration:** A code framework (e.g. terraform, ansible) must be used to document network topologies, firewall configuration, the provisioning of servers etc (everything that can be configured through different technologies).
- **Benefit:** Changing settings manually on all switches creates a human failure root cause of most financial data center outages. With Infrastructure as Code, the security policies and routing rules are applied to all nodes in parallel.

Strategy C: Energy Efficiency & Targeted PUE Strategy

- **Metric Definition:** Supports a PUE target of ≤ 1.35 annually. The PUE indicator is used to measure the whole facility power consumption relative to the power consumption of the IT equipment:
- **Strategic Execution:** Use Hot/Cold Aisle Containment, variable speed EC fans for precision cooling units, high efficiency modular UPS system (>96% efficiency in double conversion mode). Reduction in the cooling overhead due to the climate of Kathmandu.

Strategy D: Concurrent Maintainability Operational Protocol

- **Maintenance Workflows:** Operational procedures should require that a maintenance on redundant items (an additional N+1 generation, an additional N+1 chillers/PACs, two separate routes for the UPS) should be performed during normal working hours, with no fallback into emergency mode and without any interruption of any active transaction.

3.4 Comparative Analysis: Proposed Strategy vs. Standard Industry Practices

To validate the proposed strategy, the design is evaluated against standard legacy data center practices traditionally used in the region:

Evaluation Metric

Legacy/Traditional Data Center Practice

Equinix Proposed Technical Strategy

Strategic Business Advantage for eSewa

Network Architecture

Traditional 3-Tier (Core-Aggregation-Access) with Spanning Tree Protocol (STP).

2-Tier Spine-and-Leaf (Clos) Fabric with ECMP and VXLAN-EVPN.

Eliminates bandwidth bottlenecks, reduces internal latency to , and provides sub-second failover.

Redundancy & Maintenance

Tier II (capacity, single power path). Requires scheduled maintenance downtime.

Tier III Concurrently Maintainable (capacity, dual distribution paths A+B).

Guarantees zero planned service interruptions for routine hardware and utility servicing.

Expansion Capability

Monolithic deployment; scaling requires purchasing massive infrastructure upfront.

Modular Pod-based Architecture (Scale-Out).

Lower initial capital expenditure (CapEx) with the ability to scale in increments matching revenue growth.

Security Framework

Perimeter Firewall (North-South filtering only); weak internal segmentation.

Zero-Trust Micro-segmentation (East-West & North-South inspection) with HSM integration.

Strictly isolates payment databases, ensuring full PCI-DSS v4.0 compliance and preventing lateral cyber threats.

Energy Efficiency

Uncontained aisles with fixed-speed cooling; average PUE .

Hot/Cold Aisle Containment with variable-speed precision cooling; target PUE .

Reduces operational power expenditure (OpEx) by up to 25%, contributing to corporate sustainability goals.

4. Data Center Tier Standard Selection and Evaluation

4.1 Technical Classification of Uptime Institute Tier Standards

The Uptime Institute Tier Classification System is the global benchmark for data center availability, performance, and reliability. The standard categorizes data center infrastructure into four distinct Tiers based on power delivery, mechanical cooling, fault tolerance, and operational maintainability.

- **Tier I (Basic Infrastructure):** Designed for low-criticality business systems. Features a single path for power and cooling distribution with no redundant capacity components (). Susceptible to disruptions from both planned maintenance and unplanned outages.
- **Tier II (Redundant Components):** Incorporates redundant power and cooling equipment () such as extra generators or chillers, but retains a single distribution path. Maintenance on distribution lines still requires shutting down IT workloads.
- **Tier III (Concurrently Maintainable):** Features dual distribution paths (Path A and Path B) and capacity components. Every capacity component and distribution path can be isolated, serviced, or replaced on demand without causing downtime to the active IT load.

- **Tier IV (Fault Tolerant):** Built with multiple physically isolated, active distribution paths (or). Systems automatically detect, isolate, and absorb hardware or system faults without impacting the IT equipment.

4.2 Comparative Evaluation of Alternative Tier Specifications

To determine the optimal classification for eSewa's primary facility in Nepal, the four Tier levels are evaluated across critical operational and financial dimensions:

Metric / Parameter

Tier I

Tier II

Tier III (Proposed)

Tier IV

Operational Uptime SLA

Max Unplanned Downtime

Capacity Component Redundancy

(Baseline)

or

Distribution Pathing

Single Path

Single Path

Dual Path (1 Active, 1 Alternate)

Dual Active Paths

Maintenance Impact

Full Facility Shutdown

Planned System Shutdown

Zero Impact (Concurrent Maintenance)

Zero Impact

Protection Against Human Error

None

Low

Moderate to High

High (Automated Isolation)

Relative CapEx Multiplier

(Baseline)

Baseline

Relative OpEx Multiplier

(Baseline)

Baseline

4.3 Engineering and Business Justifications for the Proposed Tier

Selected Classification: Uptime Institute Tier III (Concurrently Maintainable)

In order to meet uptime institute Tier III concurrent maintainability criteria for mechanical systems, and cooling distribution, a dual loop piping header (Loop A & Loop B), plus N+1 precision In-Row CRAH units, and air cooled chillers are provided, with two isolated supply and return mains in order that a pump, isolation valve, chiller, or cooling unit can be isolated,

and worked on, without chilled airflow disruption to active server racks, or degradation of the IT thermal load.

1. Elimination of Scheduled Maintenance Outages (Why Tier III over Tier II?): Under Tier II service level, servicing up stream electrical panel, main circuit breaker or cooling header pipe should involve bringing down Computer room. For eSewa, putting core wallets service offline during schedule maintenance involves failed transactions in more than 500,000+ merchant sale outlets and appears in contradiction to the ROSA directives for the uninterrupted operation set by Nepal Rastra Bank (NRB). Tier III enables any electrical switchboard, UPS module or cooling unit to be isolated for planned maintenance so that the other route provides 100% running load.

2. Capital Efficiency & Realistic Return on Investment (Why Tier III over Tier IV?):

The availability at those tiers is also higher (99.995%) but full fault tolerant architecture (2N or 2(N+1)) necessitates duplication of all the mechanical and electrical infrastructure which increases the construction CapEx of facilities by about 85% as well as the utility due to the low utilization of electrical equipment (OpEx). Whilst a regional Payment Service Provider (PSP) based out of Nepal, Tier III is the optimum tier in terms of enterprise grade availability of infrastructure and cost of long term ownership.

1. Utility Grid Instability Mitigation:

The Kathmandu Valley host system suffers from a great deal of power quality issues and voltage fluctuation problems. Tier III configuration is a combination of N+1 Modular UPS along with N+1 standby diesel where sensitive IT equipment is separated from utility grid clean up and outages.

4.4 Risk Assessment and Redundancy Modeling ()

1. Mathematical Redundancy Formulations

In order to provide concurrent maintainability, capacity equipment is configured with N+1 redundancy which feeds two pairs of networks:

- **IT Load Capacity Requirement ():** Let represent the total active power requirement of all servers, switches, and storage arrays in the data center.

(Where *N* is the rated capacity of an individual UPS or cooling unit).

• **Tier III Component Redundancy (N):**

It is also guaranteed that should any one unit be removed for maintenance or become inoperative the remaining N units easily deliver the full load without exceeding the thermal and power limits.

• **Dual Path Distribution Modeling (Path A + Path B):**

Power is delivered via two independent, physically isolated distribution paths (Path A and Path B) terminating at dual-corded server Power Supply Units (PSUs).

If Path A fails or is isolated, Path B automatically carries 100% of the load seamlessly via the server's internal dual-corded power logic.

2. Tier III Power Distribution Architecture

3. Risk Assessment & Failure Mode Matrix

Threat / Risk Event

Failure Impact Level

Tier III Engineering Redundancy & Mitigation Strategy

Residual Risk Level

Complete Utility Grid Blackout

High

Dual utility feeds backed by local standby diesel generators with on-site fuel reserves for 48 hours of continuous operation.

Low

Primary UPS Module Failure

Critical

modular UPS architecture instantly absorbs the failed module's load; Path B continues feeding power simultaneously.

Negligible

In-Row PAC Cooling Unit Failure

Moderate

Cooling units deployed at capacity per aisle. Adjacent units automatically ramp up fan speeds to maintain temperature setpoints.

Low

Human Error During Maintenance

High

Physical key-interlock systems and strict SOPs prevent accidental isolation of active distribution paths.

Low

Fiber Cut on Primary Telecom Path

High

Geographic diversity via **Path A (NPIX Exchange)** and **Path B (Cross-Border Fiber)** ensures zero loss of internet/WAN connectivity.

Low

5. Comprehensive Data Center Network Solution System

5.1 High-Level Network Architecture: 2-Tier Spine-and-Leaf Clos Fabric

For eSewa high transactions engine, microservices API calls and real-time sync database with partner banks, the network infrastructure goes 100% off legacy 3-tier (Core-Aggregation-Access) model. To support large amounts of East-West (server-to-server) throughput and microsecond latencies of today's financial Data-Centers, legacy STP designs with redundant links blocked, cannot...

LEGACY 3-TIER VS. MODERN SPINE-AND-LEAF FABRIC

Feature

Legacy 3-Tier (STP)

Spine-and-Leaf Clos Fabric

Primary Traffic Focus

North-South (Client-Server)

East-West (Server-Server)

Loop Prevention

Spanning Tree (Blocks 50%)

Layer 3 ECMP (100% Active)

Network Latency

Unpredictable Multi-hop

Deterministic 2-Hop Latency

Scalability

Vertical (Scale-Up)

Horizontal (Scale-Out)

The Clos Network Topology Framework

The proposed network uses a **2-Tier Spine-and-Leaf topology** based on a folded 3-stage Clos network model:

1. **Spine Layer (High-Speed Backbone):** Functions as the ultra-fast interconnect creating the transit backbone between all the Leaf switches. Spine does not link to other Spines or directly to end-user servers.
2. **Leaf Layer (Top-of-Rack / Access Nodes):** Connects all physical rack servers, Storage SAN arrays, HSMs, and outer security appliances.
3. **Every-Leaf-to-Every-Spine Connectivity:** In a full-mesh bipartite graph there is a tie between all of the Leafs and all of the Spines. While this may seem as large an overhead as not having a leaf switch at all the minimum hop for any server to reach any other server in the data center is **2-hop** and deterministic.

5.2 Physical and Logical Network Topology Framework

Physical Layer Specifications

- **Spine Tier:** 2× Ultra-High-Density 400GbE Spine Switches running wire-speed switching fabrics with deep packet buffers.
- **Leaf Tier (Top-of-Rack - ToR):** Paired 25GbE/100GbE Top-of-Rack Leaf switches deployed per rack.
- **Dual-Homing Server Attachments:** Servers and storage nodes feature dual 25GbE Network Interface Cards (NICs) connected to paired Leaf switches via **Multi-Chassis Link Aggregation (MLAG)**. This eliminates single points of failure at the rack level.

5.3 Technical Justifications and Protocol Engineering

1. Underlay Network: External BGP (eBGP) Routing

- **Protocol Choice: eBGP (External Border Gateway Protocol)** is selected as the underlay routing protocol across the Spine-and-Leaf fabric.
- **Autonomous System (AS) Assignment:** Unique Private Autonomous System Number (Autonomous System Numbers 64512–65534) has been assigned to each Leaf switch pair and each Spine switch.

- **Equal-Cost Multi-Pathing (ECMP):** Using eBGP on every Leaf-to-Spine connection, ECMP uses dynamic toll-free load balancing on all the available links at the same time. If one link between a Spine switch and a Leaf is having a fiber cut, ECMP switches to other links in less than 50ms to avoid dropped financial sessions.

2. Overlay Network: VXLAN with BGP EVPN Control Plane

- **Virtual Extensible LAN (VXLAN):** The VXLAN standards define the following encapsulation; Layer 2 Ethernet frame is encapsulated within Layer 3 UDP packets (This MAC-in-UDP encapsulation fits inside any UDP packet.)... It breaks the 4,094 VLAN boundary barrier (which opens up over 16 million logical VXLAN Network Identifiers (VNIs)) and lets Layer 2 networks extend securely across a Layer 3 routed fabric.
- **BGP Ethernet VPN (EVPN):** Serves as the control plane for VXLAN that is consistent and rules-based BGP EVPN uses the MP-BGP protocol to distribute MAC and IP adjacency information throughout the VXLAN network. This eliminates broadcast storms within the datacenter by replacing the “flood-and-learn” behavior with controlled learning through a control plane.

5.4 Traffic Management, Oversubscription, and Security Policies

1. Mathematical Oversubscription Calculation

Oversubscription ratio evaluates potential bandwidth bottlenecks between downstream server connections and upstream Spine connections:

- **Leaf Switch Configuration:** 48× 25GbE Downstream Access Ports + 6× 100GbE Upstream Uplinks.
 - **Downstream Capacity:**
 - **Upstream Capacity:**
- **Calculation:**

Engineering Justification: A 2:1 oversubscription ratio is also advantageous against conventional enterprise ratios (4:1 or 6:1). It ensures that none of the internal microservice communication will encounter Queuing Congestion, even during the colossal national sales seasons, i. E. Dashain and Tihar.

2. Network Micro-Segmentation & Compliance (PCI-DSS & NRB)

- **Zero-Trust Zone Isolation:** Combining VXLAN overlay (Layer 2) with NGFW (Layer 7) effectively segments the network into isolated security zones as follows:
 - **DMZ / Public API Zone:** Handles incoming requests from mobile apps and 500,000+ QR merchant terminals.
 - **Core Application Zone:** Runs microservice container pods for wallet management and billing.
 - **Secure Database & HSM Zone:** Isolated database clusters housing encrypted user ledgers and Hardware Security Modules (HSMs) for PIN/OTP cryptography.
- **Micro-Segmentation Enforcement:** Traffic between servers in the same rack existing East-West traffic, Application, Commit/Replication traffic within the same rack are denied in compliance with Zero-Trust firewall policy. All other cross-zone traffic is denied this satisfies **PCI-DSS v4.0 Requirement 1** and **Nepal Rastra Bank IT Guidelines**.

Conclusion

Section A establishes the technical requirements for eSewa's data center based on three absolute operationally necessary features: the ability to handle tremendous holiday influx traffic such as Dashain and Tihar, compliance with the Nepal Rastra Bank regulatory requirements, and to keep the payment systems alive and processing transactions on a 24/7 basis. To do this we chose a Tier III architecture to achieve a guaranteed high availability of 99.982%, allowing planned maintenance outages for a maximum of 1.58 hours per year without affecting live systems. This relies on our modern 2-Tier Spine-and-Leaf data center network architecture running on a 400 GbE core. Through pairing eBGP routing with VXLAN-EVPN overlays, the network keeps the round-trip time for processing API calls below 50m, within the requirements of Zero-Trust micro-segmentation to achieve PCI-DSS v4.0.

Having the overall plan, reliability criterion and network mayhem, we can present the actual infrastructure. In Section B, the concrete hardware selection is given, the compute, RAM and NVMe disk capacity are sized, assuming maximum loads and directly correlated to a TIA-942 standard floor plan with rack layout.

1.2 Project Management Plan & Milestone Timeline

Project Management Methodology

Hybrid project management method is followed to ensure smooth execution, risk mitigation and close adherence to the guidelines defined of the assignment. All physical infrastructure, power and civil layout is following waterfall model to ensure strict engineering standards while software defined network and security and DCIM configuration is following Agile method for iterative design & review.

PROJECT GOVERNANCE FRAMEWORK

- Phase 1: Initiation & Business Scenario Planning (Weeks 1–2)
- Phase 2: Technical Strategy & Tier III Architecture Design (Weeks 3–4)
- Phase 3: High-Speed Spine-and-Leaf Network Engineering (Weeks 4–5)
- Phase 4: IT Hardware Selection & TIA-942 Spatial Layout Mapping (Weeks 6–8)
- Phase 5: Electrical, Cooling (MEP) & DCIM System Integration (Weeks 8–10)
- Phase 6: Final Review, APA Compliance & Submission (Weeks 11–12)

Project Milestone Timeline (Gantt Chart)

As mandated by project management standards, the following timeline outlines the key activities across a 12-week lifecycle: